

Problem 1. For any x, y positive real numbers, the geometric mean and the harmonic mean are defined by

$$G(x, y) = \sqrt{xy}, \text{ and } H(x, y) = \frac{2}{\frac{1}{x} + \frac{1}{y}}.$$

Prove that the harmonic mean is always less than or equal to the geometric mean.

Problem 2. An integer p will be called a *pumpkin* if it can be expressed as the sum of two square numbers a and b , where $a, b \in \mathbb{Z}$, i.e., $p = a^2 + b^2$.

- a) Prove that if p is a pumpkin, then $2p$ is a pumpkin.
- b) Prove that if p and q are pumpkins, then pq is a pumpkin.

Problem 3. Provide a counterexample for each statement.

- a) If pq is a pumpkin, then p or q are pumpkins.
- b) If $a, b \in \mathbb{N}$, then $a + b < ab$.
- c) If A and B are sets, then $\mathcal{P}(A) - \mathcal{P}(B) \subseteq \mathcal{P}(A - B)$.